

Nishant Choudhary

B.Tech Electronics & Computer Engineering | VIT Chennai | 2027

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EDUCATION

B.Tech – Electronics & Computer Engineering, VIT Chennai Aug 2023 – May 2027
CGPA: **8.32 / 10** Coursework: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks

TECHNICAL SKILLS

Programming	Python, C++, SQL/MySQL (Schema Design, Query Optimization, ETL)
Software Engineering	REST APIs, FastAPI, Docker, Git, GitHub, Linux (CLI), Software Architecture, Performance Optimization, pytest
AI / Machine Learning	Machine Learning, Deep Learning, Computer Vision, NLP, Transfer Learning
Frameworks	PyTorch, TensorFlow, Scikit-learn, OpenCV, Hugging Face, Streamlit
Libraries	Pandas, NumPy

SELECTED PROJECTS

ClickTutor AI – AI-Powered Desktop Learning Assistant

Python · PyQt6 · Gemini Vision API · PyTesseract · OpenCV

🐙 GitHub

- Architected a modular **MVC-based** desktop assistant integrating OCR, multimodal LLM inference, and real-time overlay rendering for context-aware explanations across any application.
- Engineered a resilient **multi-pass OCR localization pipeline** (exact, fuzzy, substring, and duplicate-aware matching), achieving **100% accuracy (116/116)** across **30 benchmark iterations**.
- Profiled the full inference pipeline (capture, OCR, LLM, localization, rendering); overlay rendering averaged **<10 ms**, while end-to-end latency was dominated by OCR and LLM inference (**~800 ms** total), guiding targeted optimisation.
- Validated the OCR localization pipeline with **pytest**, covering exact, fuzzy, and duplicate-resolution cases across **116 test samples**.

Government Budget Analytics Platform

Python · MySQL · SQL · Streamlit · ETL · Plotly

🐙 GitHub

- Designed a normalized **6-table MySQL schema** with foreign keys, SQL views, and indexed relationships for multi-year Government of India tax revenue data, ensuring referential integrity.
- Engineered a modular **ETL pipeline** (Python, MySQL) with transaction management, rollback recovery, idempotent upserts, and automated validation for repeatable ingestion.
- Optimized analytical SQL queries at **1M-row** scale, reducing complex JOIN latency by **~83% (2066 ms → 350 ms)** through targeted indexing and execution-plan analysis.
- Built a **Streamlit analytics dashboard** with KPI visuals, SQL query explorer, filtering, CSV/Excel export, and query caching, cutting dashboard response time by **~91% (420 ms → 38 ms)**.

Fake News Detection REST API

Python · FastAPI · BERT · Docker · REST API

🐙 GitHub

- Designed and deployed a **FastAPI** REST service around a fine-tuned **BERT** 6-class claim classifier (LIAR dataset), with a modular pipeline separating preprocessing, training, evaluation, and serving, plus confidence-scored predictions and structured logging.
- Containerized the service with **Docker** for production deployment; benchmarked at **8.5 ms** average / **12.8 ms** P95 latency and **396 req/sec** throughput, with a **0.67 s** cold-start model load and **418 MB** runtime footprint.

FocusLens – Real-Time Attention Monitoring System

Python · MediaPipe · Random Forest · LSTM · Streamlit

🐙 GitHub

- Engineered a real-time computer vision pipeline using MediaPipe Face Mesh, extracting **478 facial landmarks** per frame at **54.17 FPS** with zero dropped frames.
- Designed a low-latency feature-engineering stage (EAR, PERCLOS, blink rate, head-pose) running at **2.65 ms**, feeding a Random Forest + LSTM inference stage with **78.28 ms** end-to-end prediction latency.
- Achieved **97.67%** accuracy and **98.82%** F1 on evaluation data, with **96%** tracking stability across live video streams.

ACHIEVEMENTS & CERTIFICATIONS

- Solved **150+** Data Structures and Algorithms problems on LeetCode.
- Built **4 end-to-end software projects**: desktop apps, database systems, REST APIs, and AI applications.
- Earned the **Oracle OCI AI Foundations Associate** certification (2025).
- Completed Coursera: **Machine Learning Specialization** (Andrew Ng) and **Git & GitHub** Professional Certificate.